

Diferencia de medias, varianzas desconocidas e iguales $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} S_P \sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

$$S_P^2 = \frac{(n_1 - 1) S_1^2 + (n_2 - 1) S_2^2}{n_1 + n_2 - 2},$$

$t_{\alpha/2}$ tiene distribución t con $v = n_1 + n_2 - 2$ g.l.

Diferencia de medias, varianzas desconocidas y diferentes $\sigma_1^2 \neq \sigma_2^2$

$$T' = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

es aprox. una variable t con g.l.

floor function $\lfloor \cdot \rfloor$

$$\left\lfloor \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right)^2}{\left[(S_1^2/n_1)^2 / (n_1 - 1) \right] + \left[(S_2^2/n_2)^2 / (n_2 - 1) \right]} \right\rfloor$$

$$\left\lfloor \frac{35.57^2}{35} \right\rfloor$$

$$\left\lfloor \frac{17.001^2}{17} \right\rfloor$$

$$\left\lfloor \frac{59.928^2}{59} \right\rfloor$$

$$\left\lfloor \frac{35^2}{35} \right\rfloor$$

y el intervalo de confianza del $(1 - \alpha) \cdot 100\%$ es

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$s_p^2 = (n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 = (11)(0.594) + (9)(0.2) = 20$$

$$n_1 + n_2 = 20$$

Datos
 $\bar{x}_1 = 3.11$
 $s_1 = 0.771$
 $n_1 = 12$

Unknown
 0.95



$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where s_p is the pooled estimate of the population standard deviation and $t_{\alpha/2}$ is the t -value with $v = n_1 + n_2 - 2$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

0.41
 $0.59 < \mu_1 - \mu_2 < 1.54$

$n_2 = 10$
 $\bar{x}_2 = 2.04$
 $s_2 = 0.448$

Example 9.11:

The article "Macroinvertebrate Community Structure as an Indicator of Acid Mine Pollution," published in the *Journal of Environmental Pollution*, reports on an investigation undertaken in Cane Creek, Alabama, to determine the relationship between selected physiochemical parameters and different measures of macroinvertebrate community structure. One facet of the investigation was an evaluation of the effectiveness of a numerical species diversity index to indicate aquatic degradation due to acid mine drainage. Conceptually, a high index of macroinvertebrate species diversity should indicate an unstressed aquatic system, while a low diversity index should indicate a stressed aquatic system.

$\sigma_1^2 = \sigma_2^2 = \sigma^2$
 no tabenot
 evinto er
 es to
 $\alpha = 0.10$
 $\alpha/2 = 0.05, v = 20$

$$\int_{-\infty}^{t_{0.05}} h(t; 20) dt = 0.95$$

$t_{0.05} = 1.7117$

Two independent sampling stations were chosen for this study, one located downstream from the acid mine discharge point and the other located upstream. For 12 monthly samples collected at the downstream station, the species diversity index had a mean value $\bar{x}_1 = 3.11$ and a standard deviation $s_1 = 0.771$, while 10 monthly samples collected at the upstream station had a mean index value $\bar{x}_2 = 2.04$ and a standard deviation $s_2 = 0.448$. Find a 90% confidence interval for the difference between the population means for the two locations, assuming that the populations are approximately normally distributed with equal variances.

$$\text{point estimate} \pm t_{\alpha/2} \text{ s.e.}(\text{point estimate})$$

or

$$\text{point estimate} \pm z_{\alpha/2} \text{ s.e.}(\text{point estimate}).$$

For example, in the case where $\sigma_1 = \sigma_2 = \sigma$, the estimated standard error of $\bar{x}_1 - \bar{x}_2$ is $s_p \sqrt{1/n_1 + 1/n_2}$. For the case where $\sigma_1^2 \neq \sigma_2^2$,

$$\widehat{\text{s.e.}}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

Example 9.12: A study was conducted by the Department of Zoology at the Virginia Tech to estimate the difference in the amounts of the chemical orthophosphorus measured at two different stations on the James River. Orthophosphorus was measured in milligrams per liter. Fifteen samples were collected from station 1, and 12 samples were obtained from station 2. The 15 samples from station 1 had an average orthophosphorus content of 3.84 milligrams per liter and a standard deviation of 3.07 milligrams per liter, while the 12 samples from station 2 had an average content of 1.49 milligrams per liter and a standard deviation of 0.80 milligram per liter. Find a 95% confidence interval for the difference in the true average orthophosphorus contents at these two stations, assuming that the observations came from normal populations with different variances.

Solution: For station 1, we have $\bar{x}_1 = 3.84$, $s_1 = 3.07$, and $n_1 = 15$. For station 2, $\bar{x}_2 = 1.49$,



Observaciones pareadas.

→ Cada unidad experimental (lo que se mide) tiene dos posibles resultados, antes y después de algún experimento.

Dos datos no pueden ser tratados como variables aleatorias independientes.

Se quiere estudiar la diferencia de promedios: $d_1, d_2, \dots, d_n$ muestrales.

$$\mu_D = \mu_1 - \mu_2$$

$$\mu_D \rightarrow \bar{D}, \quad \bar{d}$$

$$\sigma_D^2 \rightarrow s_D^2$$

$$\bar{d} - t_{\alpha/2} \frac{s_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2} \frac{s_d}{\sqrt{n}},$$

$$V = n - 1$$

	w_b	w_a	d_i
1			
2			
3			
⋮			
⋮			
⋮			
4			

Example 9.13: A study published in *Chemosphere* reported the levels of the dioxin TCDD of 20 Massachusetts Vietnam veterans who were possibly exposed to Agent Orange. The TCDD levels in plasma and in fat tissue are listed in Table 9.1.

Find a 95% confidence interval for $\mu_1 - \mu_2$, where μ_1 and μ_2 represent the true mean TCDD levels in plasma and in fat tissue, respectively. Assume the distribution of the differences to be approximately normal.

Table 9.1: Data for Example 9.13

Veteran	TCDD Levels in Plasma	TCDD Levels in Fat Tissue	d_i	Veteran	TCDD Levels in Plasma	TCDD Levels in Fat Tissue	d_i
1	2.5	4.9	-2.4	11	6.9	7.0	-0.1
2	3.1	5.9	-2.8	12	3.3	2.9	0.4
3	2.1	4.4	-2.3	13	4.6	4.6	0.0
4	3.5	6.9	-3.4	14	1.6	1.4	0.2
5	3.1	7.0	-3.9	15	7.2	7.7	-0.5
6	1.8	4.2	-2.4	16	1.8	1.1	0.7
7	6.0	10.0	-4.0	17	20.0	11.0	9.0
8	3.0	5.5	-2.5	18	2.0	2.5	-0.5
9	36.0	41.0	-5.0	19	2.5	2.3	0.2
10	4.7	4.4	0.3	20	4.1	2.5	1.6

Primer parcial: 6 de octubre.

→ Entregar repaso a mano y en classroom.

→ Traer laptop.

→ No celular,
no tablet

→ Formulario.

→ No tablas
estadísticas

No ejercicios resueltos.

→ Máximo 11-20 entradas.

→ Módulo y aula a confirmar.

→ 1A? Entrevista oral y
entrega
a os.